Sai Teja Pegallapati

(945)-308-6251 | saitejapegallapati0520@gmail.com | Houston, TX

Associate Software Engineer

Early-career software engineer and Engineering Data Science master's student with 3+ years of experience building and supporting production-grade backend and AI-enabled systems in enterprise environments. Strong foundation in Python-based services, ML pipelines, and cloud-native development, with hands-on exposure to AI agent concepts including reasoning workflows, context handling, memory-style state management, and guardrails. Curious, fast-learning, and highly collaborative, with a strong interest in translating cutting-edge AI into secure, reliable enterprise platforms—well aligned with Proofpoint's Satori AI Agent platform and people-centric cybersecurity mission.

CORE COMPETENCIES

AI Agent Systems (Foundational) | Python Backend Development | Agent Reasoning & Workflow Orchestration | Context & Memory Handling (Foundational) | Guardrails & Alignment Concepts (Exposure) | Large Language Models (LLMs – Exposure) | Retrieval-Augmented Generation (RAG – Exposure) | Vector Search Concepts | Backend APIs & Services | Cloud-Native Architecture (Foundational) | Docker (Exposure) | CI/CD Concepts | Software Engineering Fundamentals | Debugging & Testing | ML Pipelines & Experimentation | Secure & Reliable Systems | Technical Documentation | Code Reviews | Cross-Functional Collaboration | Learning Velocity & Curiosity

PROFESSIONAL EXPERIENCE

Software Engineer

Tata Consultancy Service ***Oct 2021 – Jan 2024***

- Developed and supported Python-based backend services used in large-scale enterprise applications.
- Assisted in integrating AI/ML-driven components into production workflows with a focus on reliability, correctness, and scalability.
- Supported experimentation and evaluation of ML models, analyzing outputs, failure cases, and performance tradeoffs.
- Contributed to AI-assisted workflows involving data preprocessing, inference-style pipelines, and validation logic.
- Participated in CI/CD pipelines, deployment validation, and post-release monitoring to ensure operational readiness.
- Debugged production issues, performed root-cause analysis, and implemented fixes to reduce recurrence.
- Collaborated with platform, product, and analytics teams to translate requirements into backend features.
- Documented APIs, workflows, and operational behaviors to support long-term maintainability and knowledge sharing.

Data Science & Engineering Fundamentals

University of Houston ***Jan 2025 – Dec 2025***

- Mentored graduate students on Python programming, ML fundamentals, and AI experimentation workflows.
- Reviewed assignments involving model training, evaluation metrics, and reasoning about AI outputs.
- Assisted faculty with validating experimental results and maintaining clear technical documentation.

EDUCATION

University of Houston – *Master of Science, Engineering Data Science* ***Aug 2026***

JNTUH University – *Bachelor of Technology, Electronics & Communication Engineering* ***June 2021***

PROJECTS

Agentic AI Reasoning & Workflow Prototype – Implemented early-stage AI agent workflows using LLM frameworks to orchestrate multi-step tasks. Explored context propagation, memory-style state handling, and tool-based agent behaviors.

LLM, RAG & Retrieval Systems Exploration – Integrated LLMs with embedding-based retrieval to improve contextual relevance in responses. Experimented with vector similarity search, prompt refinement, and retrieval strategies. Analyzed hallucination patterns, failure modes, and response consistency.

Production-Style ML & AI Pipeline - Built Python-based ML pipelines for training, evaluation, and benchmarking with reproducibility in mind. Conducted controlled and ablation-style experiments to assess preprocessing and configuration impacts. Documented assumptions, results, and learnings to support iteration and collaboration.

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate
- INTRO to Machine Learning – Hewlett Packard Enterprise Data Science Institute
- Introduction to Transformer-Based Natural Language Processing – NVIDIA